Know Your Data

Data Objects and Attribute Types

- Data Sets are made of Data objects. A data object represents an entity.
- An attribute is a data field, representing a characteristic or feature of a data object.
- The type of an attribute is determined by the set of possible values – nominal, binary, ordinal and numeric.

Types of Attributes

- The following properties (operations) are typically used to describe attributes:

  1. Distinctness = and ≠
  2. Order <, ≤, > and ≥
  3. Addition + and –
  4. Multiplication * and /

Types of Attributes

- Categorical or Qualitative attributes:
  1. Nominal Attributes
  2. Ordinal Attributes
- Numeric or Quantitative Attributes:
  3. Interval scaled attributes
  4. Ratio Scaled attributes

Types of Attributes

Attribute Type		Description	Example
Categorical	Nominal	Attributes values are just different names. (Operations: $=$ and $\neq$)	Eye colour, gender.
	Ordinal	Values can be ordered. ($=, \neq, <, \leq, >, \geq$)	Grades, street number.
Numeric	Interval Scaled	Measured on a scale of equal-size units. No true zero-point. ($=, \neq, <, \leq, >, \geq$, + and −)	Calender dates, Temperature in Celsius and Fahrenheight
	Ratio Scaled	Inherent zero-point defined, ratio meaningful. ($=, \neq, <, \leq, >, \geq$, +, −, * and /)	Length, mass, age.

Basic Satistical Descriptions of Data

- **Measuring the Central Tendency**:

  Mean, Median and Mode
- **Measuring the Dispersion of Data:**

  Range, Quartiles, Variance, Variance, Standard deviation, and interquartle range, Five number summary, Boxplots and outliers
- **Data Visualization:**

  Pixel-oriented visualization, Geometric projection, Icon-based visualization, Hierarchical visualization, Visualization complex data

Measuring Data similarity and Dissimilarity

- Data Matrix versus Dissimilarity Matrix
- Proximity Measures for Nominal Attributes
- Proximity Measure for Binary Attributes
- Dissimilarity of Numeric Data
- Proximity measure for Ordinal Attributes
- Dissimilarity for Attributes of Mixed Types

Data Matrix versus Dissimilarity Matrix

- **Data objects** are described by multiple attributes. Suppose we have *n* objects described by *p* attributes. The objects are $x_1=(x_{11}, x_{12},\ldots,x_{1p})$, $x_2=(x_{21}, x_{22},\ldots,x_{2p})$ etc.
- **Data matrix** stores the n data objects in the form of *n*-by-*p* (*n*×*p*) matrix.
- **Dissimilarity matrix** stores distances between all pairs of the *n* objects. The matrix is symmetric. So, only lower triangular matrix is sufficient.

Data Matrix versus Dissimilarity Matrix

- Data matrix:

$$\begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix:

0				
d(2, 1)	0			
d(3, 1)	d(3, 2)	0		
…	…	…	0	
d(n, 1)	d(n, 2)	…	…	0

Similarity and Dissimilarity

- **The Similarity** between two objects is a numerical measure of the degree to which the two objects are alike. Similarities are usually non-negative and are often between 0 and 1.
- **The dissimilarity** between two objects is a numerical measure of the degree to which the two objects are different. Dissimilarities are lower for more similar pair of objects. Frequently, the term distance is used as synonym for dissimilarity.
- **Transformations** are often applied to convert a similarity to a dissimilarity, or vice versa or to transform a proximity measure to fall within a particular range such as [0, 1].

Similarity and Dissimilarity

- Transforming similarities to dissimilarities and vice versa is relatively straightforward. If the similarity (or dissimilarity) falls in the interval [0, 1], then the dissimlarity can be defined as :

$$d = 1 - s \qquad or \qquad s = 1 - d$$

- If the range of the measurements is not restricted to [o, 1], then transformations such as the following can be used:

$$s = \frac{1}{d+1} \qquad s = e^{-d} \qquad s = 1 - \frac{d - \min_d}{\max_d - \min_d}$$

In general, any monotonic decreasing function can be used to convert dissimilarities to similarities, or vice versa.

Similarity and Dissimilarity between Simple Attributes

- First proximity between objects having a single attribute will be discussed.

Attribute type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & if\ x = y \\ 1 & if\ x \neq y \end{cases}$	$s = \begin{cases} 1 & if\ x = y \\ 0 & if\ x \neq y \end{cases}$
Ordinal	$d = \lvert x - y \rvert / (n-1)$ Values mapped to integers 0 to n-1, *n* is the no of values	$s = 1 - d$
Interval or Ratio	$d = \lvert x - y \rvert$	$s = -d \quad s = \frac{1}{d+1} \quad s = e^{-d}$ $s = 1 - \frac{d - \min_d}{\max_d - \min_d}$

Dissimilarities between Data Objects

- The proximity of objects with a number of attributes is typically defined by combining the proximities of individual attributes.
- Various kinds of dissimilarity and similarity measures will be discussed here.

Distance Measures

- **Euclidean distance**, *d*, between two points, ***x*** and ***y***, in one, two, three or higher dimensional space, is given by the following familiar formula:

$$d(x, y) = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

where *n* is the number of dimensions and x_k and y_k are the *k*-th attributes of ***x*** and ***y***.

Distance Measures

- **Minkowski** distance generalizes the Euclidean distance:

$$d(x,y) = \left(\sum_{k=1}^{n} |x_k - y_k|^r \right)^{\frac{1}{r}}$$

Where *r* is a parameter. The following are the most common examples of Minkowiski distance.

r=1. City block (**Manhattan**, **taxicab**, **L_1 norm**) distance.

r=2. **Euclidean** distance (**L_2 norm**)

r=∞. **Supremum** (**L_{max}** or **L_∞ norm**). This is the maximum difference between any attribute of the objects.

$$d(x,y) = \lim_{n\to\infty} \left(\sum_{k=1}^{n} |x_k - y_k|^r \right)^{\frac{1}{r}} = \max_{0\le k\le n} |x_k - y_k|$$

Metric Measures

- Distances, such as the Euclidean distance, have some well-known properties. If $d(\boldsymbol{x}, \mathbf{y})$ is the distance between two points, $\boldsymbol{x}$ and $\mathbf{y}$, then the following properties hold.

1. Positivity

   (a) $d(\boldsymbol{x}, \boldsymbol{x}) \geq 0$ for all $\boldsymbol{x}$ and $\boldsymbol{y}$,

   (b) $d(\boldsymbol{x}, \boldsymbol{y}) = 0$ only if $\boldsymbol{x} = \boldsymbol{y}$.

2. Symmetry

   $d(\boldsymbol{x}, \boldsymbol{y}) = d(\boldsymbol{y}, \boldsymbol{x})$ for all $\boldsymbol{x}$ and $\boldsymbol{y}$.

3. Triangle Inequality

   $d(\boldsymbol{x}, \boldsymbol{z}) \leq d(\boldsymbol{x}, \boldsymbol{y}) + d(\boldsymbol{y}, \boldsymbol{z})$ for all points $\mathbf{x}$, $\boldsymbol{y}$ *and* $\boldsymbol{z}$.

Measures that satisfy all three properties are known as metrics.

Similarities between Data Objects

- For similarities, the triangle inequality typically does not hold, but symmetry and positivity typically do. If $s(\boldsymbol{x}, \boldsymbol{y})$ is similarity between points $\boldsymbol{x}$ and $\boldsymbol{y}$, then the typical properties are the following:
  1. $s(\boldsymbol{x}, \boldsymbol{y}) = 1$ only if $\boldsymbol{x} = \boldsymbol{y}$, $(0 \leq s \leq 1)$
  2. $s(\boldsymbol{x}, \boldsymbol{y}) = s(\boldsymbol{y}, \boldsymbol{x})$ for all $\boldsymbol{x}$ and $\boldsymbol{y}$. (Symmetry)

Examples of Proximity Measures

- **Similarity Measures for Binary Data**

  Similarity measures between objects that contain only binary attributes are called **similarity coefficients**, and typically have values between 0 and 1.

  Let ***x*** and ***y*** be two objects that consist of *n* binary attributes. The comparison of two such objects leads to the following four quantities (frequencies):

  f_{00} = the number of attributes where ***x*** is 0 and ***y*** is 0
  f_{01} = the number of attributes where ***x*** is 0 and ***y*** is 1
  f_{10} = the number of attributes where ***x*** is 1 and ***y*** is 0
  f_{11} = the number of attributes where ***x*** is 1 and ***y*** is 1

Examples of Proximity Measures

- **Simple Matching coefficient:** One commonly used similarity coefficient is the simple matching coefficient (SMC), which is defined as

$$SMC = \frac{number\ of\ matching\ attribute\ values}{number\ of\ attributes} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}}$$

- **Jaccard coefficient:** The Jaccard coefficient is frequently used to handle objects consisting of asymmetric binary attributes. The Jaccard coefficient, which is often symbolized by J, is given by the following expression:

$$J = \frac{number\ of\ matching\ presences}{number\ of\ attributes\ not\ involved\ in\ 00\ matches} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

Similarity Coefficients

- **Problem**: calculate SMC and J for the following two binary vectors.

  x=(1, 0, 0, 0, 0, 0, 1, 0, 0, 0)

  y=(0, 0, 0, 0, 0, 0, 1, 0, 0, 1)

Solution:

$$SMC = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}} = \frac{1+6}{1+2+1+6} = \frac{7}{10} = 0.7$$

$$J = \frac{f_{11}}{f_{01} + f_{10} + f_{11}} = \frac{1}{1+2+1} = \frac{1}{4} = 0.25$$